

Soumya Ranjan

Aspiring AI/ML Engineer & Data Engineer, BS in Data Science, IIT Madras
23f2004742@ds.study.iitm.ac.in · +91 7990590921 · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

Data Science undergraduate at IIT Madras and founder of an NFC/RFID self-checkout startup, seeking an entry-level AI/ML or Data Engineering role. Builds end-to-end machine learning, deep learning, and GenAI/RAG systems, from gradient-boosted regression pipelines to retrieval-augmented LLM applications and full-stack platforms (Flask, Vue.js, Node.js).

EDUCATION

IIT Madras, BS in Data Science

May 2023 - Sep 2027 (Expected)

S.E.M. High School, XII Grade (PCM)

Jun 2018 - Mar 2022

WORK EXPERIENCE

Growth Cap Partners, Software Engineering Intern

Mar - Apr 2024

- Developed software solutions for business process optimisation across internal operations.
- Collaborated with cross-functional teams to plan and implement technology solutions.

PROJECTS

Industrial Equipment & Flight Price Prediction 🐍 Python, XGBoost, LightGBM, CatBoost

- Built two tree-ensemble regression pipelines, forecasting heavy-equipment resale valuation (RMSLE) and flight ticket prices (R^2) from mixed numerical, categorical, and high-cardinality data.
- Tuned CatBoost, XGBoost, LightGBM, and Random Forest with RandomizedSearchCV and blended multi-model ensembles to improve generalisation.
- Engineered target/frequency encodings and modular preprocessing pipelines, validated with stratified cross-validation to prevent leakage.

Parts-Tray Semantic Segmentation 🐍 Python, PyTorch, OpenCV

- Built a semantic segmentation pipeline predicting pixel-level masks for 6 industrial parts classes using a custom U-Net built from scratch in PyTorch.
- Implemented a Dice-aware loss, image/mask augmentation, and test-time augmentation, encoding predictions back to RLE for submission.

Smart MCQ Solver 🐍 NLP/GenAI, Transformers, Hugging Face, RAG

- Built an AI ranking system predicting top-3 correct MCQ answers, optimised for MAP@3.
- Combined transformer-based reasoning with retrieval-augmented generation, ensembling model outputs to improve accuracy.
- Deployed the solution as an interactive Streamlit web app for real-time inference and demoing.

Bank Customer Churn Prediction 🐍 Python, XGBoost, Scikit-learn

- Built a classification pipeline predicting customer churn, benchmarking Logistic Regression, Random Forest, Gradient Boosting, and XGBoost, optimised for F1-score on imbalanced data.
- Engineered class-balanced training and a custom decision-threshold sweep (vs. default 0.5 cutoff) to maximise F1.

IFC File Error Detection System 🐍 Python, Scikit-learn, Random Forest, Solo Project

- Built an ML-based error detection system for IFC files used in BIM workflows, automating structural and syntactic validation.
- Engineered features from IFC schema hierarchies and implemented a Random Forest classifier from scratch for error-type prediction.
- Improved classification accuracy through hyperparameter tuning, reducing manual review effort for BIM engineers.

College Placement Portal 🐍 Vue.js, Node.js, Flask, SQLAlchemy, Redis, JWT

- Built a role-based placement platform (students, recruiters, admins) with a Vue.js SPA, JWT auth/RBAC, and Flask + Node.js services.
- Integrated Celery/Redis for async notifications; designed relational models and REST APIs for applications, interviews, and offers.
- Implemented GraphQL endpoints alongside REST for flexible, client-driven queries across the platform.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, SQL
ML	Scikit-learn, CatBoost, XGBoost, LightGBM, Random Forest, PyTorch, Pandas, NumPy
NLP & GenAI	Transformers, Hugging Face, RAG, Prompt Engineering, LLM Fine-Tuning, ANN, RNN, DNN
Backend & Web	Flask, SQLAlchemy, REST APIs, GraphQL, JWT, Celery, Redis, Node.js, Vue.js
Databases & Cloud	PostgreSQL, MySQL, AWS, Docker, Linux, Git, CI/CD
Visualisation	Matplotlib, Seaborn, Power BI, Tableau

LEADERSHIP & ACHIEVEMENTS

- Founder & CEO, SoApp**, NFC/RFID self-checkout startup for organised retail; incubated at Nirmaan (IIT Madras), patent application under review.
- Treasurer & former Society Affairs Head, Shah Maat** (IIT Madras BS Chess Club), managed finances for a 500+ member organisation.
- Diploma in Programming, IIT Madras**, coursework in Application Development, Data Structures & Algorithms, Database Management Systems, and System Commands.
- Diploma in Data Science, IIT Madras**, coursework in Machine Learning, Deep Learning, Generative AI, and Data Science tools.